

SilentWhisper: faint whisper speech using wearable microphone

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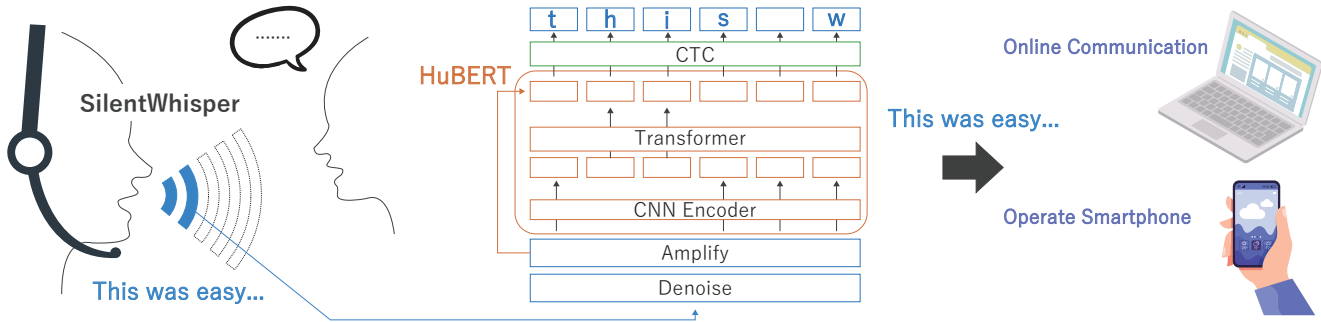


Figure 1: An overview of SilentWhisper system. The user performs voice input using whispers that cannot be recognized by others. The acquired voice is subjected to background noise removal, amplified, and then subjected to speech recognition.

ABSTRACT

Voice interaction is a fundamental human capacity, and we can use voice user interfaces just speaking. However, in public spaces, we are hesitant to use them because of consideration for their surroundings and low privacy. Silent speech, a method that recognizes the movement of speech in silence, has been proposed as a solution to this problem, and it allows us to maintain our privacy when speaking. However, existing silent speech interfaces are burdensome because the sensor must be kept in contact with the face and mouth, and commands must be prepared for each user. In this study, we propose a method to input whispered speech at a quiet volume that cannot be heard by others using a pin microphone. Experimental results show that a recognition rate was 13.9% WER and 6.4% CER for 210 phrases. We showed that privacy-preserving vocal input is possible by whispering voices which are not comprehensible to others.

CCS CONCEPTS

• Human-centered computing → Gestural input.

KEYWORDS

whisper, murmur, silent speech, speech, voice user interface

ACM Reference Format:

Hiroataka Hiraki and Jun Rekimoto. 2022. SilentWhisper: faint whisper speech using wearable microphone. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3558715>

1 INTRODUCTION

Voice user interface is one of the basic communication means and can be used hands-free and eye-free. In recent years, they have been used in smartphones, smart devices, and car navigation systems. On the other hand, voice user interfaces are often not used even in environments where they are actually available because they can be a nuisance to those around them and because they do not maintain privacy. For example, Aarathi[15, 16] found in a study that users are cautious about transmitting personal information in situations where they use voice assistants on smartphones Rico et al.[20] investigate the importance of small, unobtrusive, and non-attentive methods for gesture input. Therefore, it is important to have an input method that does not reveal the input to the surroundings.

A silent speech interface[3] has been proposed as a means to solve this problem. Silent speech interface is a method to recognize voice commands by measuring tongue and lip movements during silent speech. In particular, silent speech using sound has been proposed by Non Audible Murmur(NAM)[17] that acquire faint whispers from a microphone attached to the skin, and by using a throat microphone. While these methods can acquire only the user's own voice, they generate spike noise when the users moves their body. Another input method using breathing has also been proposed, but it is difficult to input while breathing in. Then,

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UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

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ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558715>

Pretrained Model	train	test	WER(%)	CER(%)
HuBERT with librispeech-960	wTIMIT(W)	wTIMIT(W)	0.30	0.14
	wTIMIT(W), CSTR	SilentWhisper	22.6	10.7
	wTIMIT(W), CSTR, SilentWhisper	SilentWhisper	13.9	6.4

Table 1: Results of whisper speech recognition. We used a HuBERT[7] pretrained on librispeech-960h[18]. For finetuning, train and test columns are as follows. wTIMIT(W):whisper dataset in wTIMIT corpus. CSTR:whisper speech dataset recorded with headset in clean and noisy environment.

we propose SilentWhisper, which acquires whispers that are not comprehensible to the surroundings using a pin microphone. Experimental results showed that recognition rate was 13.9% WER and 6.4% CER for 210 phrases. This result is better than the existing NAM recognition rate of 81.2%[12] and can be used as input interface. In this study, we showed that privacy-preserving input using vocal input is possible by whispering voices which are not comprehensible to others.

2 RELATED WORK

Various methods have been proposed for silent speech, including myoelectricity[8], ultrasonic[11], lip-reading[10, 19, 21], touch sensor[9, 13], and others. Here, we focus on silent speech methods using sound.

Nakajima et al. have proposed a method called NAM[17], in which the sound passing through the body is acquired by meat conduction. This method can use the sound inside the body as input, and it is possible to provide input that is not audible to the surroundings. However, microphone must be kept in contact with the user’s body, so spike noise occurs when the user moves their body.

Other methods have been proposed that use a speaker and microphone to acquire mouth movements from acoustic reflections[5, 22, 23]. These are non-contact, but are not suitable for hands-free configurations because the direction of input is fixed or the user must hold the smartphone.

Fukumoto[4] proposes a method that uses inhalation breathing, which is input by sound of breath inhalation. This method is difficult to use because it requires the user to learn how to speak while inhaling, although the sound is not leaked to the surroundings because the input is made while inhaling.

3 SILENTWHISPER

3.1 Dataset for Training

wTIMIT:. wTIMIT(whisper TIMIT)[14] is natural and whispered speech corpus with 450 phoneme-balanced sentences from the TIMIT corpus in two accents(Singaporean-English, and North American). This corpus is spoken by somewhat gender-balanced 20-28 speakers. We divided the training and testing data as already separated. Preliminary experiments showed that the word error rate(WER) dropped by about 10% when Singaporean-English was recognized in North American English, so we only use North American dataset.

CSTR NAM TIMIT Plus: CSTR NAM TIMIT Plus(CSTR)[?] includes a parallel whispered speech recorded simultaneously via a non-audible murmur (NAM) microphone and an omni-directional headset. The corpus comprises 421 sentences selected from newspaper, 460 sentences selected TIMIT corpus, and 18 isolated words. The CSTR corpus using headphone was recorded in two situations: clean(76 minutes) and noisy(81 minutes), which are 157 minutes in total.

SilentWhisper: We selected 421 sentences from TIMIT corpus, which are the same as the CSTR corpus. We collected these 421 sentences including 1154 words with directional pin microphone attached to the neck of the garment from 1 user. The collected data was 42 minutes in total and each files was resampled to 16000Hz and denoised with wiener filter.

3.2 Recognition Architecture

For recognizing SilentWhisper, we used HuBERT[7], a self-supervised learning based speech recognition. A HuBERT model is pre-trained on large unlabeled speech data and then fine-tuned by SilentWhisper dataset using Connectionist Temporal Classification(CTC)[6]. For tokenizing CTC outputs into character sequences, we used Wav2Vec2.0[2] Tokenizer. Before fine-tuning the model, the sampling frequency of the datasets was set to 16000Hz. When fine-tuning, the batch size was 32 and 1000 epochs were turned using a single GPU:NVIDIA A100.

3.3 Results

The recognition results for SilentWhisper are shown in Table 1. The existing whispering dataset, wTIMIT is recognized with high accuracy by fine-tuning. On the other hand, when evaluating SilentWhisper using wTIMIT and CSTR, the accuracy is not too high. This is thought to be because the volume of SilentWhisper’s voice is small and considered as noise. Therefore, we used 20 minutes of SilentWhisper’s voice as training data for fine-tuning, and obtained 13.9% WER and 6.4% CER. In addition, amplifying the whole volume to match the wTIMIT corpus improves the recognition results, we obtained 13.

4 DISCUSSION AND CONCLUSION

We proposed SilentWhisper, a whisper that cannot be comprehensible to others, and performed recognition with a pin microphone. As a result, 100 phrases could be input with a recognition rate was 13.9% WER and 6.4% CER for 210 phrases. In the future, the recognition rate and vocabulary can be increased by converting existing natural speech datasets to SilentWhisper. In addition, recognition

at the phoneme level rather than the character level and the introduction of language models will enable more accurate recognition.

ACKNOWLEDGMENTS

This work was supported by JST [Moonshot R&D][Grant Number JPMJMS2012]

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